

- Why is MLFlow useful?
MLFlow simplifies managing the lifecycle of machine learning models by providing tools for tracking experiments, packaging code, and deploying models.
- What is the purpose of having automated pipelines in ML production environment?
Automated pipelines ensure reproducibility, scalability, and efficiency, making it easier to deploy, test, and maintain ML models in production. Relying on manual notebooks introduces inconsistency and error risks.
- What is the difference between A/B testing, Green Blue deployments, Canary deployments, and Challenger Champion deployments?
- A/B Testing: Compares two versions of a model to see which performs better.
- Green/Blue Deployments: One version (Blue) is live while Green is tested, then switched.
- Canary Deployments: Gradual rollout of new versions to a small group before full deployment.
- Challenger/Champion: One model (Champion) is live, and a new model (Challenger) is tested in parallel.
- Why are we hosting artifacts in AWS S3 and metadata in RDS?
AWS S3 provides scalable and secure storage for large artifacts, while RDS offers reliable, structured storage for metadata, enabling efficient querying and management.
- What are some security concerns and challenges involved with using secrets and API keys to authenticate our github actions workflow instance?
Secrets and API keys can be exposed if not properly managed, risking unauthorized access. Challenges include securing credentials, rotating them regularly, and preventing accidental leaks in logs or code.
- What was the biggest challenge for you in completing the assignment?
The biggest challenge was ensuring seamless integration between various tools (MLFlow, AWS, GitHub) and managing dependencies across environments.

Repository

<https://github.com/Karolina-Kuska/ARISA-MLOps.git>